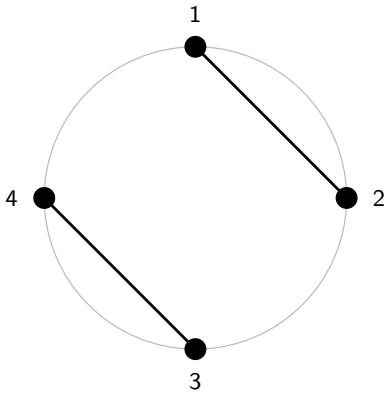
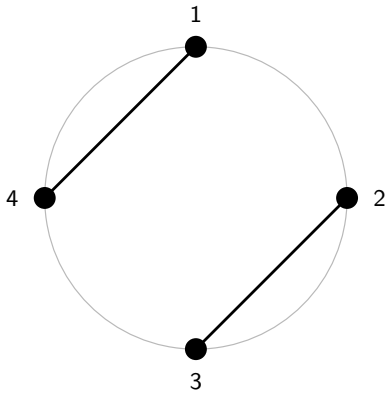
Week 12 / Catalan bijections / K–1

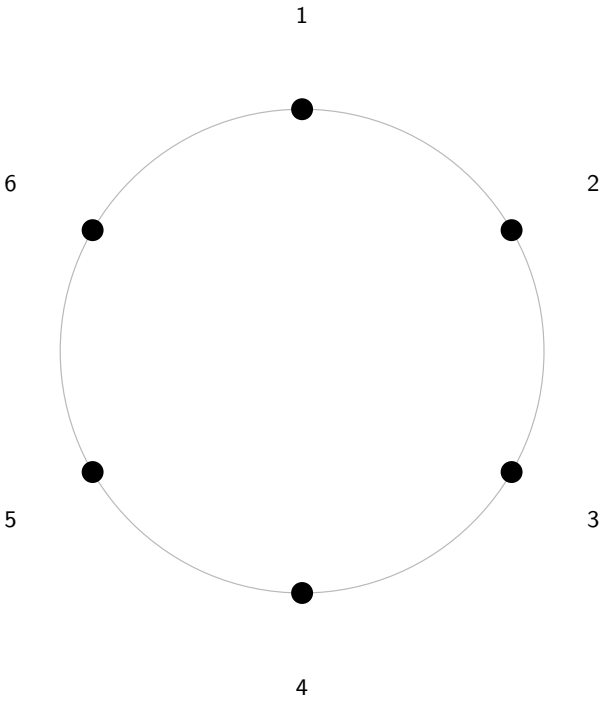
Use each dot once. Keep dots in place. Strings stay inside the circle without crossings. Pairings are different when a dot has a different partner. Use strings on large circles and a pencil on small circles.

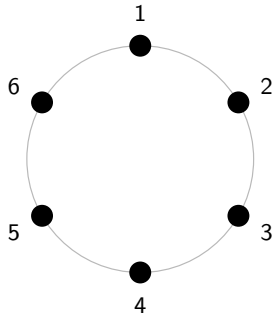
The two pairings of four dots:

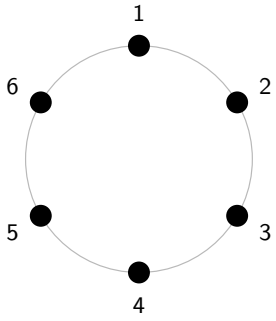


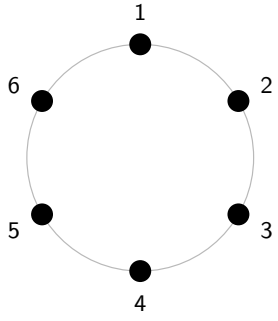


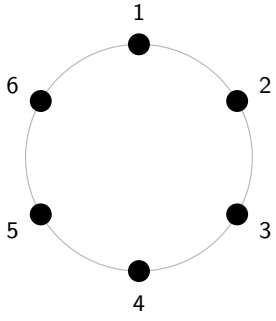
Problem 1: Find every way to pair six dots.

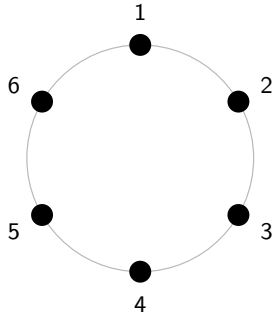


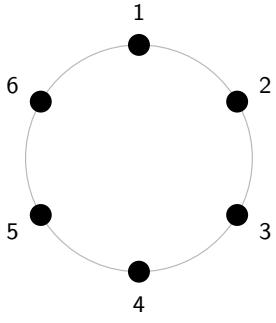




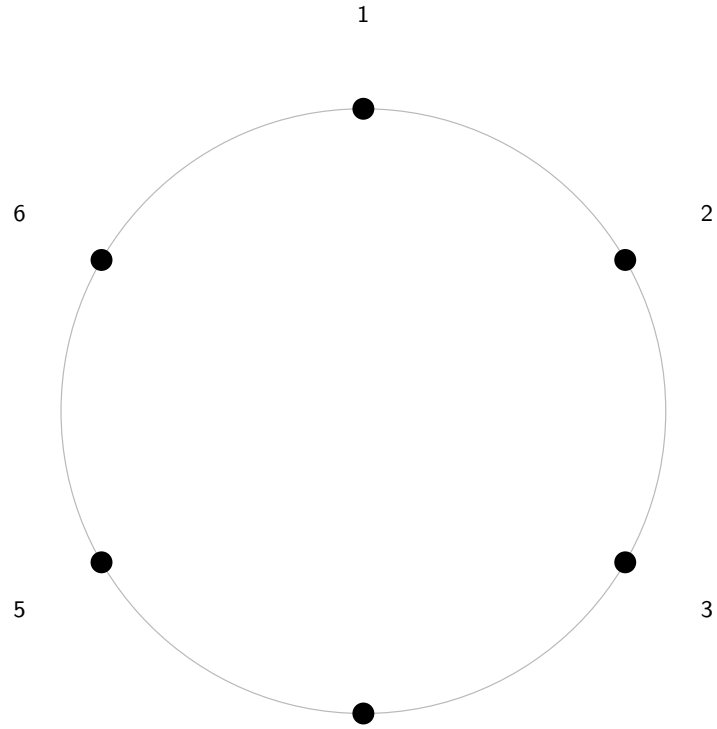


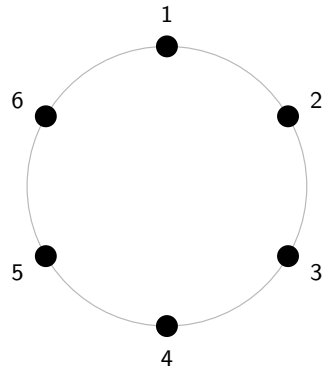


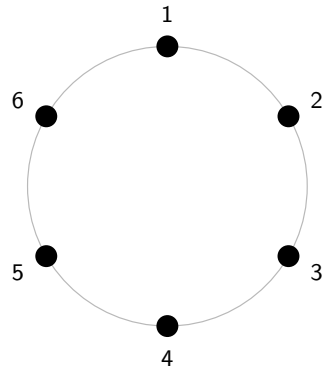


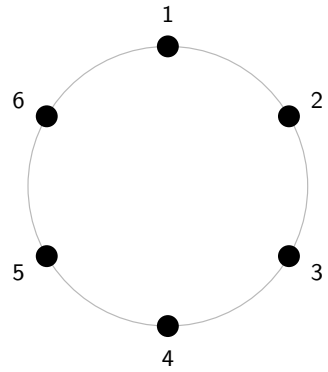


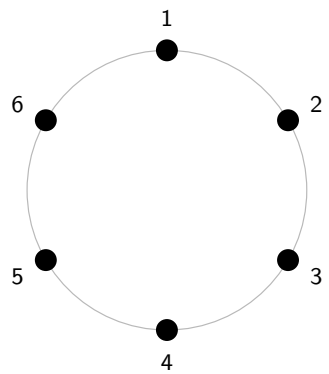
Problem 2: Check your six-dot collection. Show how you know that no pairing is missing.

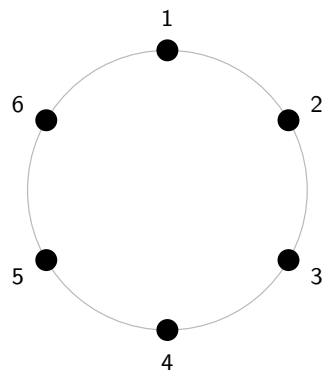


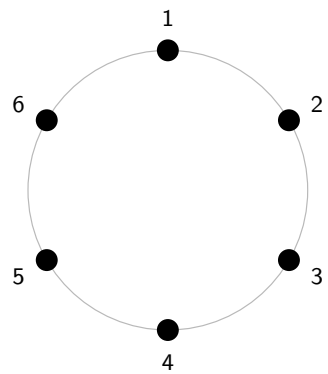




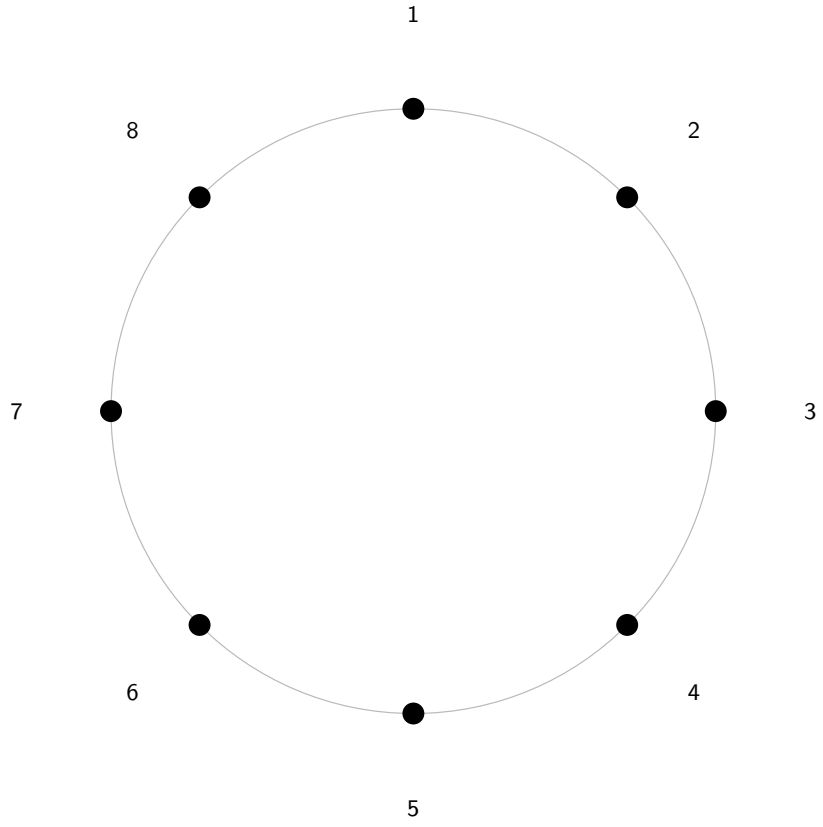


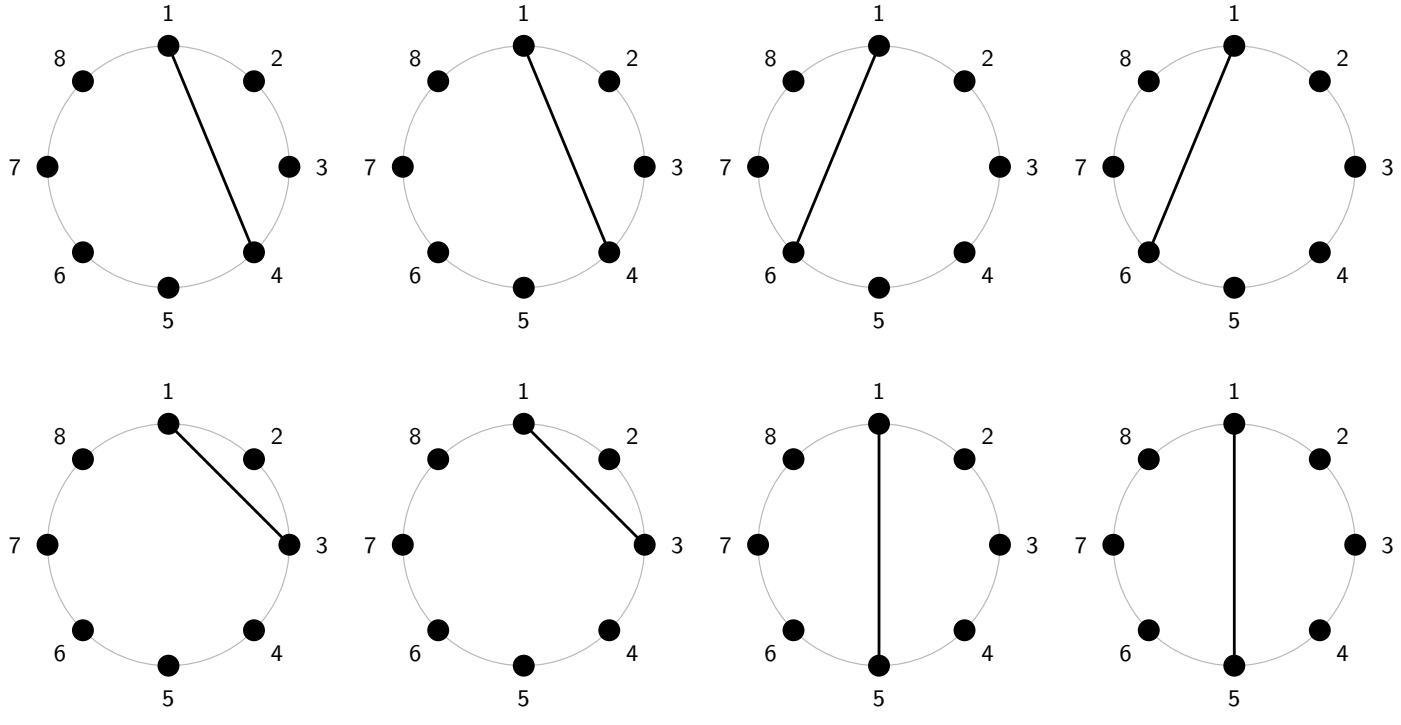






Problem 3: For each drawn pair, put a string on the large circle and leave it there. Find every way to finish the pairing, or show why you cannot.





Problem 5: Pair all the dots in each circle if you can. If you cannot, show what gets in the way.

